NOT MEASUREMENT SENSITIVE

MIL-T-46791D
11 September 1991
SUPERSEDING
MIL-T-46791C
29 November 1983

MILITARY SPECIFICATION

TRAILERS, CARGO: 500, 2,000 AND 3,000 POUND PAYLOAD, 2 WHEEL, MODIFIED COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two-wheel, commercial, cargo trailers having load-carrying capacities of 500 pounds, 2,000 pounds and 3,000 pounds under normal operating conditions. Vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in acquisition documents.

1.2 Classification. The vehicle shall be one of the following types and sizes, as specified (see 6.2).

Type I - 1/4-ton, 500-pound payload.
Type II - 1-ton, 2,000-pound payload.
  Size A - Body inside dimensions:
    48 inches wide (minimum).
    96 inches long (minimum).
  Size B - Body inside dimensions:
    72 inches wide (minimum).
    110 inches long (minimum).
Type III - 1-1/2 ton, 3,000-pound payload.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2330

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS
MILITARY

*	MIL-B-46176	- Brake Fluid, Silicone, Automotive and All Weather, Operational and Preservative.

STANDARDS
FEDERAL

*	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking, and Data Plate Standards.
*	MS 51336	- Lunette-Coupler, Drawbar, Ring.
*	MS 51339	- Coupler, Drawbar Ring: Offset (Tapered Shank), 60,000 Lb. GVW.
*	MS 75020	- Connector, Plug, Electrical - 12-Contact, Intervehicular, 28-Volt, Waterproof.
	MS 75021	- Connector, Receptacle, Electrical: 12-Contact, Intervehicular, 28-Volt, Waterproof.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

AMERICAN NATIONAL STANDARD INSTITUTE, INC. (ANSI)
American National Standard Pipe Thread Series.

(Application for copies of ANSI publications should be addressed to the American National Standard Institute, 1430 Broadway, New York, NY 10018.)

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* DEPARTMENT OF DEFENSE
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DoT publications should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of the ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SAE, INC.
SAE Standards and Recommended Practices

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD adopted).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J588	- Turn Signal Lamps.
J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
J702	- Brake and Electrical Connection Locations - Truck-Tractor and Trailer (DoD adopted).
J844	- Non-metallic Air Brake System Tubing (DoD adopted).
J1067	- Seven-Conductor Jacketed Cable for Truck-Trailer Connections.
J1292	- Automobile Truck, Truck-Tractor, Trailer and Motor Coach Wiring.
J1401	- Hydraulic Brake Hose - Automotive.

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J1402 - Automotive Air Brake Hose and Hose Assemblies.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

(Application for copies of Tire and Rim Association publications should be addressed to The Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.5, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items which meet or exceed the requirements of this specification. All items shall be as represented in the manufacturer's technical data book. Special or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to delivery of the items.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

* 3.1.1.2 Markings and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223. When specified (see 6.2), concealed markings shall be furnished in accordance with 4.1.3 of MIL-STD-1223.

* 3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

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3.1.1.4 Wood treatment. Wood shall be treated in accordance with MIL-STD-1223.

3.1.1.5 Silicone brake fluid. When specified (see 6.2), and if available as the manufacturer's standard or optional brake fluid, brake fluid conforming to MIL-B-46176 shall be provided in the hydraulic brake system. A decal or plate shall be placed near the master cylinder stating "CAUTION: USE SILICONE BRAKE FLUID ONLY, MIL-B-46176."

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

* 3.2.2 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

3.2.3 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity. Minimum payload shall conform to 1.2 for the specified size of the vehicle.

* 3.2.4 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.5 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

3.3 Performance. Vehicle performance as specified in 3.3.1 through 3.3.5 shall be met by the trailer when fully equipped with all attachments and accessories, with payload uniformly distributed over trailer load area, completely serviced, and towed by a towing vehicle over improved and unimproved, hard, uneven terrain. Vehicle shall show no evidence of failure or malfunction of trailer, trailer components, attachments, and accessories.

3.3.1 Towing speed and trailing ability. The trailer shall follow the towing vehicle over dry, level, smooth, prepared, hard-surfaced roads at sustained speeds up to 55 miles per hour (mph), and over dry, unimproved and gravel roads at speeds up to 25 mph. The trailer shall follow the towing vehicle without exceeding the tracking deviations specified in Federal Motor Carrier Safety Regulation 393.70(a).

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3.3.2 Turning ability. When coupled to a towing vehicle operating in its minimum turning circle, the trailer shall follow without cramping. The trailer shall be capable of moving through an angle of not less than 60 degrees in relation to a 96-inch wide towing vehicle without interference between vehicles. The angle shall be measured between the longitudinal axis of the trailer and the longitudinal axis of the towing vehicle at the lunette.

3.3.3 Suspension system. The axle, springs, accessories and components of the suspension system shall withstand the imposed loads without evidence of permanent set, part failure or malfunctioning under the operating conditions specified.

* 3.3.4 Service brakes. The service brakes of type II and type III trailers, in conjunction with a towing vehicle (truck-trailer combination), under all conditions of loading, shall control, decelerate and stop the vehicle combination in accordance with Federal Motor Carrier Safety Regulation 393.52. Brake linings shall be of nonasbestos material.

3.3.4.1 Automatic actuation. The automatic breakaway devices for type II and type III trailers shall apply and maintain application of the trailer service brakes for not less than 15 minutes upon breakaway from a towing vehicle and shall conform to Federal Motor Carrier Safety Regulation 393.43(d).

3.3.5 Parking brakes. The parking brakes shall hold the trailer with specified payload on a dry, hard surfaced grade of not less than 30 percent when uncoupled from the towing vehicle, resting on the trailer's running gear and landing leg, and headed up and down the incline. The parking brake lever shall have sufficient moment arm to require not more than 30 pounds of effort for application and release. The parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41.

3.4 Payload. Minimum payload for type I, type II and type III trailers shall be 500 pounds, 2,000 pounds, and 3,000 pounds respectively. Requirements shall be met with the payload uniformly distributed over the load space.

3.4.1 Weight distribution. With the longitudinal axis of the trailer horizontal, the center of gravity of the trailer, without payload and with a uniformly distributed, full, rated payload, shall be forward of the axle. The maximum vertical load on the lunette, with the trailer unloaded, shall be 100 pounds for type I and 150 pounds for types II and III.

3.4.2 Gross weight. Gross weight shall consist of the weight of the chassis and body with all attachments and the payload. The gross weight of type I trailer shall be not more than 3,000 pounds.

3.5 Dimensions. The trailer, when uncoupled from the towing vehicle and without payload, on level ground, shall conform to the dimensions specified in table I (see 3.3.2 for clearance between towing vehicle and trailer).

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TABLE I. Dimensions.

Characteristic	Type I	Type II		Type III
		Size A	Size B	
Overall length	Minimum practicable	Minimum practicable	170 inches (max.)	160 inches (max.)
Overall width	60 inches (max.)	80 inches (max.)	96 inches (max.)	96 inches (max.)
Overall height (without racks)	48 inches (max.)	60 inches (max.)	60 inches (max.)	60 inches (max.)
Body, inside: Length Width	 72 inches (min.) 41 inches (min.)	 96 inches (min.) 48 inches (min.)	 110 inches (min.) 72 inches (min.)	 110 inches (min.) 75 inches (min.)
Height of metal panels and tailgate	12 inches (min.)	18 inches (min.)	18 inches (min.)	18 inches (min.)
Floor height	30 inches (max.)	35 inches (max.)	35 inches (max.)	35 inches (max.)
Ground clearance	9 inches (min.)	9 inches (min.)	9 inches (min.)	9 inches (min.)

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3.6 Frame. The trailer shall be equipped with a frame of all-steel, electrically welded construction, with each frame member in one piece without splices or joints. There shall be not less than nine full width crossmembers, including ends and stub crossmembers as required to provide rigidity. Crossmembers shall be of full channel, pressed or rolled steel construction reinforced by gusset plates or brackets at points of attachment to longitudinal sills. Longitudinal sills shall be reinforced. Crossmembers, longitudinal sills and reinforcements shall be fabricated of not less than 12 gage (.1046 inch) carbon steel or 13 gage (.0897 inch) high tensile steel, in accordance with U.S. standard gage sizes. The sprung weight of the loaded vehicle shall not impose a fiber stress greater than 50 percent of the yield strength of the materials used under mobile operating conditions specified herein (see 3.3).

3.7 Drawbar and coupling device. The trailer shall be equipped with a drawbar of the "A" frame type formed by the continuation of the main frameside rails. The coupling device shall be an offset-ring drawbar coupler conforming to MS 51336 and MS 51339 or other design providing a 4-inch vertical range of adjustment. The coupler shall be positioned such that the longitudinal axes of the coupler and the trailer are parallel with the trailer empty, level, and resting on the landing leg. With the trailer level and the coupling device in the higher position, the distance from the ground to the centerline of the coupling device shall be 26 inches, plus or minus 1 inch for type I trailers and 34 inches, plus or minus one inch for type II and type III trailers.

3.7.1 Landing leg. Type I trailer shall have a retractable landing leg. Types II and III trailers shall have a retractable landing leg with caster type landing wheel and a permanently installed cranking handle (rigid or swivel type) located on the curb side. The landing leg and related parts shall withstand without permanent deformation the combined static and dynamic forces produced by coupling and uncoupling operations. The landing leg shall operate freely and shall be protected from malfunctioning caused by foreign matter. The wheel shall be not less than 6 inches in diameter and shall be fitted with an antifriction bearing. Means shall be provided to lubricate the landing leg and wheel. When placed in travel position, the landing leg shall remain positively locked to provide ground clearance in the course of normal towing operations.

3.7.2 Safety chains. The trailer shall be equipped with two safety chains with eyes. The chains shall conform to Federal Motor Carrier Safety Regulation 393.70(d).

3.8 Suspension. The trailer shall be equipped with springs of the semielliptical type. The suspension system shall have a rated capacity at least equal to the load imposed on each member, measured at the ground, with the trailer loaded with the specified payload.

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3.9 Axle. The trailer shall be provided with a single axle. The axle rating shall be at least equal to the load imposed on the axle, measured at the ground, with vehicle loaded with the specified payload.

* 3.10 Wheels, rims, tires and tubes. The trailer shall be equipped with single wheels and tires. The rim size and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations. Multi-piece rims shall not be furnished. Rims and disc wheel sizes shall be the same for all wheels on each vehicle. Tire size and load range (ply rating) shall be the same for all tires on the vehicle.

* 3.10.1 Tires. Tires shall be of the tubeless type with highway tread. Tires shall be of the steel belted radial type. Unless otherwise specified, standard profile tires shall be furnished. When specified (see 6.2), low profile tires shall be furnished. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded with the specified payload. Tires shall conform to the Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

* 3.10.2 Inner tubes. When tube type tires are specified in acquisition documents, inner tubes shall be of the heavy duty type, and shall be of proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

* 3.10.3 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material.

* 3.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished. The spare wheel or rim shall be the same size as the wheels in use on the vehicle.

3.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, load range and tread design as the tires installed on the vehicle.

3.11 Brakes.

3.11.1 Services brakes (type II and type III). Unless inertia brakes or air-over-hydraulic brakes are specified (see 3.11.2 and 3.411.3), type II and type III trailers shall be furnished with a 12-volt electric brake system wired in accordance with 3.12.4. The system shall include a manual controller kit for mounting on the dash of the towing vehicle, a controller

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kit for synchronizing the trailer electric brakes with the brakes of the towing vehicle and an emergency breakaway kit. Kits shall be complete with all necessary parts (except storage battery for breakaway kit) and installation instructions. Unless otherwise specified, the synchronizing controller shall be for use with hydraulic-brake-equipped towing vehicles. When specified (see 6.2), the synchronizing controller shall be for use with air-brake-equipped towing vehicles. The kits shall be boxed, marked as to contents and securely attached to the trailer for shipment.

3.11.2 Inertia brakes (type II and type III). When specified (see 6.2), type II and type III trailers shall be furnished with an inertia (surge) applied hydraulic brake system. The inertia brake actuator, mounted on the tow bar, shall include a master cylinder, an accumulator chamber and a hold-off spring to prevent brake applications due to road shock or trailer pitching. The system shall provide for modulated brake applications, shall be protected against the entrance of foreign matter, shall include breakaway features conforming to Federal Motor Carrier Safety Regulation 393.43(d), and shall enable the trailer to be backed up without continued application of the trailer brakes.

3.11.3 Air hydraulic brakes. When specified (see 6.2), type II and type III trailers shall be equipped with air-over-hydraulic brakes. Torque ratings of the brakes, components and accessories of the brake system shall be at least equal to the respective imposed loads under the operating conditions specified herein. The service brakes shall be adjusted so that when brakes are in the released position, the brake shoes shall not be in contact with the brake drums and the minimum clearance for new brake shoe linings is provided. Brake assembly manufacturer's specification for new brake shoe clearance shall be on file at the contractor's plant and shall be provided with each service manual. With hydraulic system free of air, and with both air lines connected to compressed air source, the air system shall be stabilized at rated working pressure. Gage pressure drop shall not exceed 1 pound per square inch (psi) per minute with brakes released, nor 2 psi per minute with brakes normally applied. Hydraulic fluid leakage shall not be permitted and air leakage shall not occur at the air chamber of the power cylinder. Location of air fittings shall be in accordance with SAE J702. The braking system shall be installed in a manner which provides road clearance for travel over uneven terrain and protection against damage caused by objects striking components. No part of the braking system shall extend below the bottom of wheel rims. Hydraulic brake hoses shall be provided with a means of protection from road hazards. The braking system, complete with all necessary components, shall include at least the following equipment:

(a) Air reservoir, having not less than 1,000 cubic inches capacity and each tank equipped with readily accessible drain valve
(b) Bleed-down relay emergency valve
(c) Air powered master cylinder

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(d) Air connectors with SAE J318 gladhand couplers for emergency and service lines
(e) Air brake tubing and pipe in accordance with SAE J844
(f) Hydraulic brake hose in accordance with SAE J1401
(g) Permanent identification tags for emergency and service lines
(h) Dummy gladhand couplers with retaining chains.

3.11.3.1 Air hose assemblies. Two air hose assemblies shall be furnished with an hydraulic brake vehicle. The hoses shall be packaged and secured to the trailer for shipment. Air hoses shall be in accordance with SAE J1402, equipped with SAE J318 gladhand couplers at each end. Air hoses shall be provided with hooks or fastening devices to secure hoses adjacent to trailer frame when not in use. Each hose shall be not less than three feet longer than the distance from the trailer air connectors to the lunette.

3.11.4 Parking brakes. Manually applied parking brakes shall be furnished on all trailer types. The parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41. Provisions shall be made for slack adjustment. When brake cables are used, they shall be of corrosion-resistant steel. Lubrication fittings shall be provided.

3.11.5 Wheel chocks. When specified (see 6.2), four wheel chocks shall be provided. Wheel chocks shall be secured to the trailer with corrosion-resistant plated chains. Provisions for stowage in their respective areas shall be provided.

* 3.12 Lighting. A 12-volt direct current (dc) lighting system shall be provided. All vehicle lights, reflectors, and wiring (including electric brakes) shall conform to Federal Motor Carrier Safety Regulation 393.14 (type II, size B and type III trailers), 393.15 (type I and type II, size A trailers), 393.19, 393.20, 393.22, 393.23, 393.25, 393.26 (a, b, c and d), 393.27, 393.28, 393.29, 393.32 and 393.33. Lights and reflectors shall not be mounted on vertical surfaces of the rub rails (unless recessed and fully protected) or vehicle bumpers. All wiring shall conform to SAE J1292. The brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other.

3.12.1 Electrical receptacle. A trailer electrical receptacle, conforming to SAE J560, shall be installed on the forward end of the drawbar. The receptacle shall have a spring loaded cover assembly. Conductors shall be connected and color coded as specified in SAE J560.

3.12.2 Electrical jumper cable. A 7-conductor electrical jumper cable conforming to SAE J1067, with a cable plug at each end conforming to SAE J560, shall be provided. Both plugs shall be equipped with a grip for withdrawing from connector sockets. The cable shall be not less than three feet longer than the distance from the trailer electrical receptacle to the lunette. Stowage provisions having an anti-theft feature and means of hanging the cable, with the plugs pointing downward, shall be provided.

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3.12.3 24-volt receptacle system. When specified (see 6.2), the trailer shall be equipped with one 12-contact receptacle with cover and one 7-contact receptacle with cover installed at the front of the trailer. The 12-contact receptacle shall conform to MS 75021-1. The 7-contact receptacle shall conform to SAE J560 round socket for jumper cable. The 12-contact receptacle shall be provided with resistance to each circuit to reduce the voltage of a tactical Military Design towing vehicle from nominal 24 volts to 12 volts. The 7-contact socket shall be wired in accordance with SAE J560 to supply nominal 12 volts directly to the trailer circuits. The 24-volt 12-contact receptacle shall be connected through the necessary resistors into the 12-volt trailer circuits as follows:

(A) Do not connect.
(B) Connect to yellow circuit.
(C) Do not connect.
(D) Connect to white circuit.
(E) Connect to black and brown circuits.
(F) Do not connect.
(H) Do not connect.
(J) Connect to green circuit.
(K) Connect to blue circuit (if auxiliary circuit is used in trailer wiring).
(L) Connect to white circuit.
(M) Connect to trailer electric brakes, if furnished.
(N) Connect to trailer electric brakes, if furnished.

Circuits B and J on tactical trucks are combination stop and turn indicator circuits. The normal 12-volt turn signal lights will function both as turn signals and stop lights and the normal 12-volt stop lights vehicle with a 24-volt power supply. Because of this condition the stop light red circuit is not connected to the 24-volt 12-contact receptacle.

* 3.12.3.1 Resistor assembly. Each circuit resistor shall be selected to reduce the 24-volt dc regulated voltage of the tactical truck to within the maximum rated voltage of the trailer electrical components. The resistor assembly shall be located in a protective housing and provided with adequate ventilation or a heat sink to prevent overheating and any damage to resistors, wiring or adjacent components. A pulse type voltage reducer is acceptable.

3.12.3.2 Twelve-conductor jumper cable. A twelve-conductor jumper cable shall be furnished for each trailer furnished with a 24-volt receptacle system. The cable shall be equipped with a connector conforming to MS 75020-1 at one end and a connector conforming to MS 75020-2 at the other end. The cable shall be not less than three feet longer than the distance from the trailer electrical receptacle to the lunette. Stowage provisions having an anti-theft feature and means for hanging the cable, with the plugs pointing downward, shall be provided.

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3.12.4 Brake wiring. Electric brakes shall be wired to the auxiliary circuit of the SAE J560 seven-conductor electrical connector. All wiring shall be properly mounted and secured with clamps. Wiring shall be heavy wall thermoplastic insulated cable and shall be enclosed in conduit.

3.13 Bumper. A rear bumper shall be furnished and shall conform to Federal Motor Carrier Safety Regulation 393.86. The bumper shall extend not less than three inches beyond the rear of the trailer cargo body.

3.14 Fenders. The vehicle shall have fenders over the wheels. The fenders shall have inside shields and shall have sufficient crown to prevent mud and water from splashing on chassis and trailer body.

* 3.14.1 Rear wheel splash and stone throw protection. The rear wheels shall have mud flaps at the rear. A metal strip not less than 0.125 inch thick and not less than 1 inch wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1.250 inch wide by 0.094 inch thick shall be furnished at each bolt. All splash shield and mud flap installations, shall conform to the rear wheel splash and stone throw protection provisions of SAE J682.

3.15 Lubrication. All moving parts of the equipment normally requiring lubrication shall be provided with suitable means for such lubrication. The lubrication fittings shall be of the standard commercial type with a surface check-ball.

3.16 Body. The body shall be cargo type, having metal sides with stake pockets and a tailgate, and the dimensions as specified in table I. The body shall be of all-metal welded construction. All welds shall be continuous. Drain holes shall be provided in the front of the body floor. Body side panels shall be flanged outward into a box section at the top. Stake pockets, extending the full height of the body sides, shall be provided. The stake pockets shall be equally spaced and shall be arranged as follows: one in the center of the front and rear panels, four per side for type I and five per side for types II and III. Drainage shall be provided in each stake pocket. Tiedown devices for securing a tarpaulin shall be provided.

3.16.1 Material. Trailer body shall be fabricated of steel. Floor and tailgate shall be not less than 10 gage (.1345 inch) carbon steel. Sides, front panels, stake pockets and posts shall be not less than 14 gage (.0747 inch) carbon steel. High tensile steel may be furnished in one gage lighter weight.

* 3.16.2 Tailgate. The tailgate shall be hinged at the bottom and provided with fabric, rubber or plastic covered stay chains of a length to retain the tailgate in a horizontal position. Rear lights and reflectors shall be positioned to prevent damage when the tailgate is lowered.

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3.16.3 Side and end racks. Full width front and rear and full length side removable rack sections shall be provided. Racks shall be equipped with locking devices to lock each rack section to the body. Body hardware attaching bolts shall be not less than 2 inches from the end of the rack slats. The front and rear racks shall fasten to the side racks by not less than two removable pins on each side. Rack slats shall be of oak, hickory, or ash of not less than 0.750 inch finished thickness. The height of the racks shall be not less than 16 inches for type I, and 28 inches for types II and III, measured up from the top of the metal panels. There shall be not less than four slats per rack. Width of individual slats shall be manufacturer's standard, providing that total of slat widths is not less than 60 percent of total rack height. Slat edges shall be beveled, and end corners shall have top and bottom of slat ends rounded with not less than 0.500 inch minimum radius. Slats shall be riveted or bolted to the inside of upright posts, with the rivet or bolt heads against the slats. Upright posts shall extend into all body stake pockets the full height of the body side. Stake posts shall be of box section construction with vertical openings for bows of cross sectional dimensions of 1 x 2.500 inches.

3.16.4 Body mounting. Unless the body is integrally mounted and welded to the subframe, the body shall be mounted with not less than three U-bolts or twin studs per side, having a minimum of 0.500 inch body diameter. Tie plates shall be at least 0.500 inch thick. Two shear bolts shall be provided, one on each side of the rear portion of the body subframe, to maintain body alinement on the trailer chassis.

* 3.16.5 Tarpaulin and bows. When specified (see 6.2), the trailer shall be furnished with bows, tarpaulin, and tiedowns. Four bows for type I and five bows for types II and III shall be furnished. The bows shall be of the knockdown type constructed of metal or hardwood and metal components and shall provide an inside height of not less than 45 inches for type I and 60 inches for types II and III, measured from the floor to inside of the tarpaulin at the center. The tarpaulin shall cover the body, with bows in place, and extend down the sides, front, and rear to within 3-inches of the platform. The front and rear tarpaulins shall be the roll-up type. Tarpaulin shall be fabricated from laminated vinyl nylon or vinyl polyester material not less than 18 ounces weight per square yard. The tarpaulin shall be treated to be resistant to fire, water, weather, and mildew, and all seams shall be double stitched or heat sealed. Tiedowns shall be solid rubber straps with S-hooks at ends. Reinforcing pads, of the size, shape, and in the position required to protect the tarpaulin at the rack and panel corners, and roof bows, shall be sewn or heat sealed to the tarpaulin.

3.17 Servicing and adjusting. Prior to acceptance of the vehicle by the Government, the contractor shall service and adjust each vehicle for operational use including at least the following: alinement of wheels, inflation of all tires, proper adjustment of the brake system, and complete lubrication of chassis and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point.

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3.18 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry. All weldments shall be thoroughly cleaned of welding slag and spatter prior to painting.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine the curb weight and the distribution of curb weight on the lunette and the axle. The loads imposed on the lunette and the axle with the

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specified payload will be calculated to ascertain that the suspension, axles and tires furnished are of adequate capacity to meet contractual requirements.

4.3.2 Road test. To determine conformance to 3.3.1 through 3.3.5, the first production vehicle shall be road tested, with and without payload, by the contractor, to assure the vehicle will operate in accordance with contractual requirements. Unless otherwise specified (see 6.2), the road test shall be for a distance of not less than 500 miles. Half the mileage shall be accomplished with the trailer fully loaded and half shall be with the trailer empty. Approximately half the mileage shall be on gravel or dirt roads at speeds up to those deemed safe. At least 5 sudden stops shall be made from a speed of not less than 20 mph to determine conformance to 3.3.4 through 3.3.5. Tracking ability shall be observed to verify conformance to 3.3.1. Turning ability shall be observed to verify conformance to 3.3.2. After the test, the trailer shall be examined for evidence of misalinement, binding or other malfunction. The test course shall be arranged so as to return the vehicle to the manufacturer's plant at the completion of the test for reinspection and chassis servicing. All costs of the test, including all costs for the towing vehicle, exclusive of personal expenses incurred by participating Government representatives, shall be borne by the contractor.

4.3.3 Electrical test. An electrical test shall be conducted on the first production vehicle to determine that all vehicle lights and the electric brakes, when furnished, are functioning. In addition, when a 24-volt receptacle system is furnished, the circuits of the 12-contact receptacle shall be energized with not less than 28 volts dc regulated voltage for a time sufficient to stabilize the temperature of the resistor assembly for not less than 30 minutes in an ambient temperature of not less than 77 degrees Fahrenheit. After the test the resistor assembly, including resistors, wiring and adjacent components, shall be examined for evidence of overheating, deterioration or damage.

* 4.3.4 Test failure. The vehicle shall successfully complete the entire test. Failure of the vehicle to successfully complete the test shall be cause for nonacceptance of the contract quantity, pending correction of deficiencies and evidence of corrective action which shall prevent the recurrence of similar deficiencies in the contract quantity. Failure of the vehicle to successfully complete the test shall not constitute an excusable delay in meeting scheduled deliveries. The equipment shall be closely monitored throughout the test for interference with towing operations, and for instability as evidenced by poor tracking or uncontrollable sideslipping, skidding, swerving, or tilting. The equipment also shall be closely monitored for loosening of parts, interference between parts, leakage of fluids or lubricants, overheating of components, and for damage, permanent distortion, or excessive wear of parts and components. The equipment shall be disassembled to the extent necessary for inspection for unusual wear or damage to components. Substitution of a new trailer or replacement of a

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major component may require complete retest at the discretion of the Government representatives. Rejection of the test trailer shall be for damage or deficiencies, including but not limited to the following:

(a) Damage caused by collision
(b) Failure of any major component
(c) Vibration due to misalinement of wheels or frame
(d) Vibration due to the type of construction or mounting
(e) Evidence of abnormal tire wear due to misalinement or unbalance
(f) Failure of any vehicular safety device such as brakes or electrical circuits
(g) Evidence of structural weakness in any part of the vehicles, vehicle components, accessories or welds
(h) Loose mountings of parts or accessories due to workmanship or vehicular operation.

4.3.5 Refurbishing of test vehicle. Upon successful completion of the road test and final examination, the vehicle shall be cleaned, serviced and refurbished to the extent required for delivery of a new vehicle. Warranty provision of the contract shall be exclusive of total test mileage.

4.3.6 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.7 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.6 has been accomplished.

5. PACKAGING

5.1 Vehicle processing. Vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with manufacturer's standard commercial practice.

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6. NOTES

6.1 Intended use. The trailers covered by this specification are commercial units intended for general purpose nontactical cargo handling.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and size of trailer required (see 1.2).
* (c) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Identification of appropriate service for painting (see 3.1.1.1).
(e) Identification of appropriate service for marking (see 3.1.1.2).
(f) Rustproofing, if required (see 3.1.1.3).
(g) Tropical rustproofing, if required (see 3.1.1.3).
(h) Silicone brake fluid, if required (see 3.1.1.5).
* (i) A spare tire carrier, if required (see. 3.10.3).
(j) A spare wheel or rim, if required (see 3.10.4).
(k) A spare tire assembly, if required (see 3.10.5).
(l) An air brake synchronizing controller for electric brakes, if required in lieu of hydraulic brake synchronizing controller (see 3.11.1).
(m) Inertia brakes, if required in lieu of electric brakes (see 3.11.2).
(n) Air-hydraulic type brakes, if required in lieu of electric brakes (see 3.11.3).
(o) Wheel chocks, if required (see 3.11.5).
(p) Twenty-four volt receptacle system, if required (see 3.12.3).
(q) Tarpaulin and bows, if required (see 3.16.5).
(r) Road test mileage, if not 500 miles (see 4.3.2).

* 6.3 Subject term (key word) listing.

Commercial
Lunette
Trailer
Two wheel.

6.4 Identification of changes. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER (Number) AND TITLE MIL-T-46791D TRAILER, CARGO: 500, 2,000 AND 3,000 POUND PAYLOAD, 2-WHEEL, COMMERCIAL

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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